OSMANIA UNIVERSITY

FACULTY OF ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)

B.E. (AIML/CSE) V- Sem (Main) Examinations June-2026

Subject: Data Base Management System

Time : 1.5 hours **Max. Marks : 30**

Note : *i) Answer **Question No. 1** (Compulsory) and Answer any **two questions** from the remaining questions (2- 4), each question carrries 10 Marks..*

*iii) **Missing data, if any, may suitably be assumed.***

		Marks	BT	CO
1. a)	Define the following terms (i)Entity (ii) Relationship Set (iii) Key constraints (iv)Participation Constraints	2	1	1
b)	What are the advantages of Database Management System.	2	1	1
c)	Differentiate between Two tier and three tier Client Server Architectures.	2	1	1
d)	Define the following terms: relation schema and relational database schema, with suitable examples.	2	1	1
e)	Explain the following terms with suitable examples: Candidate Key, Primary Key and Super Key.	2	1	1
2. a)	Write notes on Generalization, Specialization and Aggregation relations with suitable examples	5	3	2
b)	Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.	5	3	2
3. a)	Explain all the variations of JOIN operation.	5	3	2
b)	Consider the following Company Schema (Fig 2) and answer the following queries using Relational Algebra a) Select the employee tuples whose department number is 4 or salary is greater than 30000. b) Select the tuples for all employees who either work in department 4 and make over 25,000 per year, or work in department 5 and make over 30,000 c) list each employee's first and last name and salary d) retrieve the first name, last name, and salary of all employees who work in department number 5.	5	3	2
4. a)	Briefly describe the following (Any Two) (i) Parts of basic SQL query (ii) SQL Commands (iii) SQL Data types and Literals (iv) SQL Operations	5	3	2

b)	Consider the University Schema as shown in figure 2 and answer the following queries using SQL a) Find the average salary of instructors in each department b) Find total sales after 10/1/2005 per product. c) Find the names and average salaries of all departments whose average salary is greater than 45000 d) Find courses offered in Fall 2017 and in Spring 2018 e) Find courses offered in Fall 2017 but not in Spring 2018	5	3	2

Fig 1

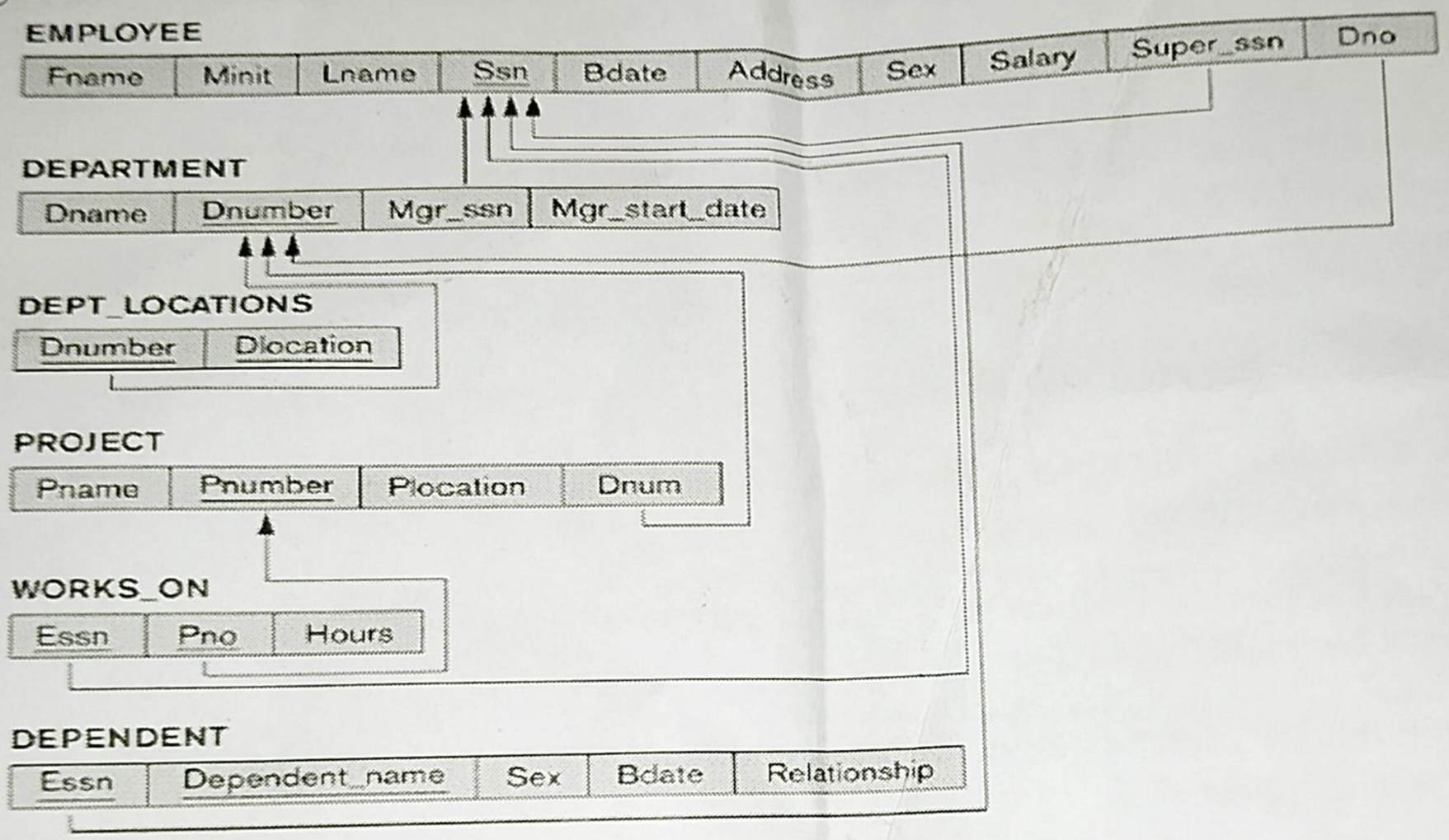


Fig 2

